PosePro: Real-Time Form Intelligence

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Abstract

PosePro is an AI-powered system that uses a webcam to analyze exercise form in real time. It delivers instant feedback, detects errors with over 94% accuracy, and assigns a movement score (0–100), helping users prioritize quality over quantity for safer, more effective workouts.

Objective

Build a real-time, software-only system that analyzes human movement during resistance exercises, detects common form errors such as shrugging, swinging, and limited range of motion with ≥90% accuracy, and delivers a dynamic weighted movement score to standardize performance evaluation and reduce injury risk by promoting correct technique.

Methodology

PosePro uses MediaPipe Pose (33-point tracking) and OpenCV to run at 30+ FPS, calculating joint angles and detecting errors like:

- Shrugging (>3 cm)
- Swinging (>1.2× velocity)
- Limited range (50°–90°)

A state machine ensures accurate rep counting (99.2%).
It scores form using weighted metrics:

- Range of Motion (25%)
- Symmetry (20%)
- Stability (20%)
- Alignment (20%)

- Smoothness (15%)

Built with a Python (Flask) backend, SQLite, and a web dashboard, the system runs with <200 ms latency.

Results

PosePro scores each repetition using five metrics: range of motion, symmetry, stability, alignment, and smoothness, combined into a 0–100 score with <200 ms latency.

In a pilot study (n=15):

- Scores improved from 62.4 to 82.1 (+31.7%)
- Inter-arm deviation reduced from 12.3° to 4.1°
- Jerky motion decreased by 58%

User feedback:

- 87% preferred score-based feedback
- 93% reported higher motivation

These results indicate improved technique and engagement.

Conclusion

PosePro shifts fitness from repetition counting to movement-quality intelligence. By adding a competitive scoring layer and delivering sub-200 ms feedback without hardware, it makes expert coaching accessible anywhere. Scalable across gyms, clinics, and home use, PosePro sets a new standard for how technique is measured, improved, and engaged with.

References

- Google. (2021). MediaPipe Pose.
- OpenCV. (n.d.). Open Source Computer Vision Library.

Acknowledgement

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Presented By

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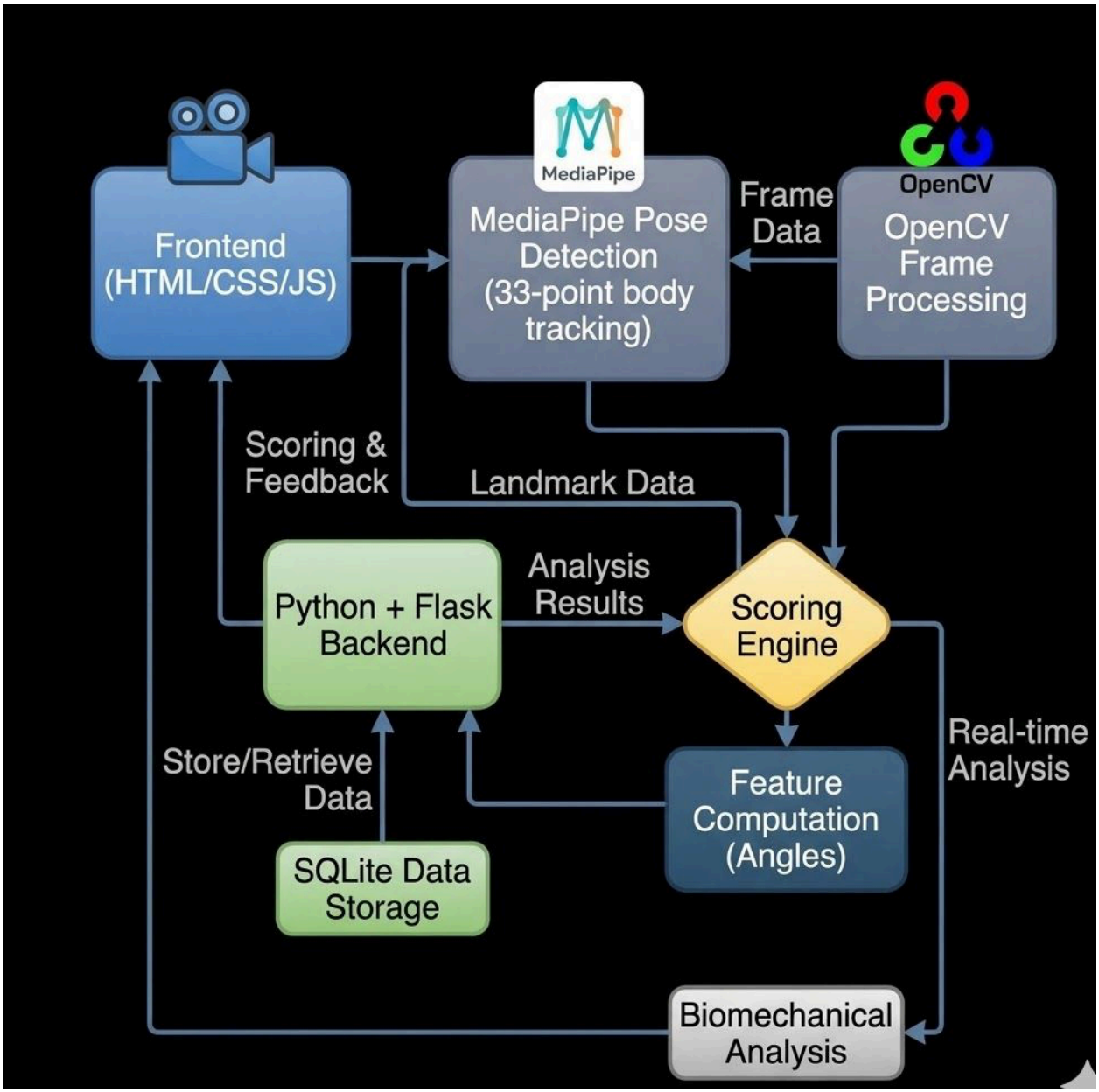
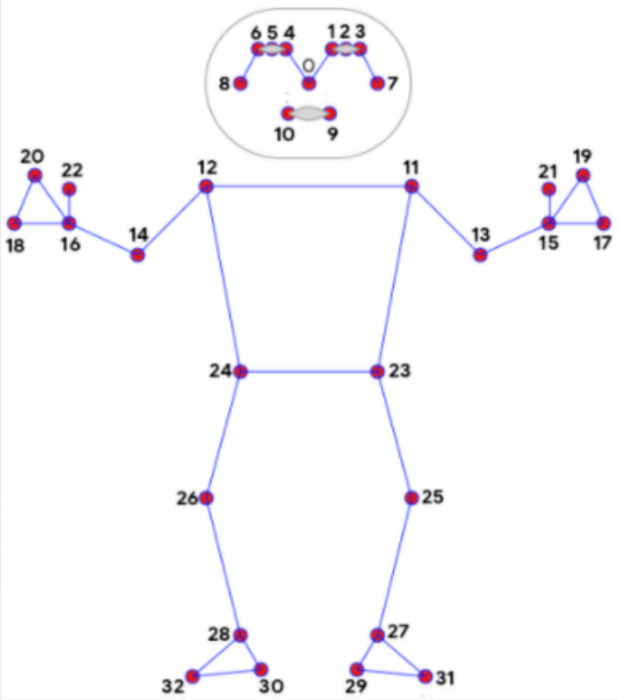
Diagrams


Frontend
(HTML/CSS/JS)
MediaPipe
MediaPipe Pose
Detection
(33-point body
tracking)
Frame
Data
OpenCV
OpenCV
Frame
Processing
Scoring &
Feedback
Landmark Data
Python + Flask
Backend
Analysis
Results
Scoring
Engine
Real-time
Analysis
Store/Retrieve
Data
Feature
Computation
(Angles)
SQLite Data
Storage
Biomechanical
Analysis

PosePro
AI-POWERED FITNESS ANALYZER

Mediapipe Landmarks



0. nose
1. left_eye_inner
2. left_eye
3. left_eye_outer
4. right_eye_inner
5. right_eye
6. right_eye_outer
7. left_ear
8. right_ear
9. mouth_left
10. mouth_right
11. left_shoulder
12. right_shoulder
13. left_elbow
14. right_elbow
15. left_wrist
16. right_wrist
17. left_pinky
18. right_pinky
19. left_index
20. right_index
21. left_thumb
22. right_thumb
23. left_hip
24. right_hip
25. left_knee
26. right_knee
27. left_ankle
28. right_ankle
29. left_heel
30. right_heel
31. left_foot_index
32. right_foot_index

Image: **https://miro.medium.com/v2/resize:fit:1400/0*6CVJq3zw2_qb9Nj4**